

Cognitive Kernel Networks (CKN)

Geometric Dominance and Preservation of Computation

John P. Alioto

Protocol v1.4.2 - Release Candidate

September 5, 2026

Abstract

Cognitive Kernel Networks (CKN) seek to increase resistance to adversarial displacement of privileged conditioning while retaining coherence and task-relevant capability comparable to a declared baseline. Stronger dominance may impose costs in generality or capability; collapse of the computation is not a successful tradeoff. This paper gives a precise sufficient theory for one form of that objective: preserving the task operation selected by privileged initialization. A geometric region is connected to the actual task transition and emitted output. A local restoration-versus-disturbance inequality keeps execution in that region, and local task-matching identities imply preservation of complete reference task traces. An abstract construction satisfies both conditions for arbitrary declared task transducers while permitting nonzero input coupling. A separate shared-state example uses the same observable for task state and protected geometry. A numerical instance exhibits an override in an unrestored comparator and its prevention in the protected system. These are mathematical results about specified computations. They establish neither a neural implementation nor general natural-language instruction authority. Weight construction, achievable language-model capability tradeoffs, and empirical performance remain unresolved.

1. Objective and scope

Model builders control the weights governing inference and can establish privileged conditioning before admitting subsequent input. CKN asks whether the induced dynamics can sustain that conditioning while supporting input-dependent computation. Moving first is an initialization advantage; persistence requires a property of the subsequent dynamics.

CKN seeks to increase resistance to adversarial displacement of privileged conditioning while retaining coherence and task-relevant capability comparable to a declared baseline. Stronger privileged dominance constrains how inputs can redirect computation and may therefore reduce generality or impose capability costs. The objective is to improve resistance to override while preserving substantive generative competence. Achieving dominance through systematic incoherence or collapse of task capability does not satisfy that objective.

All admitted inputs may perturb the geometry. Privilege does not require inaccessible embedding directions, zero coupling, unchanged working state, or concealed representations. The privileged mother tongue and its embeddings may be public. The question is whether their privileged initialization establishes a dynamical regime that later inputs cannot displace.

This paper develops a sufficient theory for **authority over a specified task operation**: the initialized configuration selects which declared task transition and output law governs the computation. Subsequent inputs may change the task data and state but cannot change that specified law on the declared domain. The operation is a formal object, not a score for acceptable answers.

The theory does not automatically identify such operations in a trained model or translate arbitrary natural-language instructions into them. This is a limitation of the formalization, separate from the question of obtaining suitable weights. A claim about general prompt-defined authority would require an additional connection that this paper does not establish.

1.1. Mathematical commitments

Version 1.3.0 required a declared margin to stay positive and gave a valid comparison of reference margin with total margin loss. Its central interpretation remained underdefined: a preserved scalar could be unrelated to the computation it purported to protect. Its global disturbance hypothesis also carried most of the desired trajectory conclusion.

Version 1.4.0 adds three obligations:

- **An operational meaning:** the geometric region must be connected to specified task transitions and actual emitted observations.
- **A dynamical implication:** assumptions on individual transitions must imply persistence over the complete interaction.
- **Joint non-vacuity:** a single mathematical system must preserve the identified computation and resist displacement despite nonzero input coupling.

The results below meet those obligations for the stated operation-authority formalization. This release candidate retains the proofs and shared-state witness, distinguishes admitted-input responsiveness from decoder variability, and explicitly permits endogenous privileged-token stabilization. It clarifies the case in which geometry factors through task state and gives the optional capability comparison globally defined baseline bounds. The paper includes the complete arguments needed to assess its claims. It does not establish a general solution to the broader CKN objective.

2. Why span exclusion is insufficient

Let the embedding image be $U = \text{im}(W_E)$. A vector outside U cannot equal a linear combination of embedding columns. That statement does not constrain subsequent influence. In two dimensions, $U = \text{span}(e_1)$ together with $Ae_1 = e_2$ permits an input entering the first direction to affect the second after one transition.

Nor does being outside a span mean being orthogonal to it: $e_1 + e_2$ is outside $\text{span}(e_1)$ but has a nonzero component along e_1 . A separated entry representation does not establish a bound on coupling through the dynamics.

Likewise, a direction annihilated by the current readout can become visible after a later transformation. A bound on each step's displacement permits accumulated drift. Fixed topology or curvature does not identify which operation controls the computation. None of these observations establishes privileged persistence.

The replacement is a dynamical condition on a geometrically identified regime, together with a separate, explicit connection from that regime to the task operation it protects.

3. Running state and declared interaction

Let $\Sigma \in \mathcal{S}$ be the complete running state needed to continue computation, including relevant context, recurrent state, and decoder-side state. For fixed parameters Θ , write

$$\Sigma_{t+1} = F_{\Theta}(\Sigma_t, e_t), \quad y_{t+1} = Y_{\Theta}(\Sigma_t, e_t).$$

Write an event as $e_t = (a_t, \omega_t)$, where a_t is admitted task input and ω_t supplies any exogenous randomness or declared decoder controls needed to make the transition deterministic. Deterministic examples can omit ω_t . A declared no-new-input value permits internal steps between arrivals. Sampled tokens are produced by the transition and decoder; they are not externally fixed output choices. With state and event fixed, the transition and emitted task observation are deterministic. The readout may depend on the updated state through its definition; it is not assumed to occur before the transition physically.

3.1. Initialization and endogenous stabilization

Let $p \in \mathcal{P}$ identify the privileged configuration and set

$$\Sigma_0 = \text{Init}_{\Theta}(p).$$

The running map does not receive a newly authorized external p at each step. This does not prohibit the model from generating, regenerating, or re-entering privileged tokens during inference to sustain the initialized regime. Such endogenous re-expression, including a privileged-token reasoning continuation, belongs to the running dynamics. Privileged representations may be public; the formal role being protected is their role in the initialized computation. Internal provenance or token spelling alone does not establish that role.

The complete state and transition must include the effects of generated tokens, their re-entry, and the resulting later computation. The geometric inequality must hold over this complete recurrent process. Thus endogenous self-stabilization is permitted by the theory; whether weights can sustain it in a useful model remains open. No stabilization mechanism or training procedure is prescribed.

Pathwise quantification covers every declared realization of decoder randomness, including realizations on which particular stabilizing tokens are not emitted. Their absence need not itself violate the bound; the full transition must still satisfy it. Probabilistic success in emitting those tokens does not establish this every-realization condition.

The theorem compares transitions at a declared common granularity. Additional internal steps can be included in the state-transition model, but differing clocks or step counts do not acquire a trace-equivalence guarantee without an appropriate comparison. Authorized replacement of initialization or weights defines a different run.

3.2. Admitted domain and integrity assumptions

Provided input is not necessarily admitted context. Validation, normalization, tokenization, and interface limits can determine what reaches the model. The paper takes the resulting admitted domain as given; it does not require an identity mapping from a submitted request to context, nor prescribe an admission pipeline. Responsiveness is assessed on admitted task input.

For the main results, fix an event alphabet $\mathcal{E}$ and quantify over every finite history in $\mathcal{E}^*$. This rectangular domain is an explicit, potentially conservative scope choice. It includes arbitrary repetitions and adaptive

choices within the alphabet. An admitted event can have a fixed size limit and the running state can have bounded context. Quantifying over arbitrary history length does not require admitting unlimited tokens into context at once. Finite context capacity alone does not specify which initialized information is retained or removed. A more restricted history domain needs a closure argument; it cannot be obtained by deleting histories that defeat the claim. Stopping can be represented by a declared terminal event and continuation rule.

Public copies of privileged symbols belong to the event domain when admitted by the interface. Knowledge of those symbols or their embeddings need not be restricted. A public symbol's effect depends on the complete running state and its place in the transition. A later copy can oppose the initialized regime while remaining insufficient to displace it. Any interface distinction required by a particular realization belongs to that realization's declared assumptions.

The model assumes the integrity of weights, initialization, and execution: the admitted-input actor cannot modify them through infrastructure access. Knowing a representation is not the same operation as replacing Θ or Init_Θ .

These boundary assumptions define the interaction under study. A token or context limit does not by itself prove the dynamical bound: finite inputs may have effects that accumulate through later inference. All admitted effects, including feedback through self-generated tokens, remain covered by the one-transition obligation below.

The main theorems are pathwise: they hold for every declared event sequence. Equality of stochastic output laws additionally requires a common law for the exogenous randomness driving the declared decoding policy, or another explicitly justified probability-preserving coupling, under the same prescribed input experiment. Matching some possible decoder realization is insufficient.

4. What privileged conditioning governs

4.1. A declared task operation

Specify a family of reference task transducers before assessing the candidate system:

$$x_{t+1} = B_p(x_t, e_t), \quad y_{t+1}^p = H_p(x_t, e_t), \quad x_0 = x_0^p.$$

Here $x \in \mathcal{X}$ is task state and $y \in \mathcal{Y}$ is the actual task observation being claimed. The maps may represent stateful computations and arbitrary finite emitted strings. They need not be linear, contractive, or differentiable. The result makes no assumption about training objectives.

The specification must identify the task operation represented by (B_p, H_p) and the effect that changing p has on that operation. Choosing the whole candidate model as its own reference after inspecting its trajectories would not supply an independent authority specification. A reference obtained from a named baseline must be fixed before the comparison; it cannot simply be labeled privileged because it was initially prompted with p .

The reference is a mathematical comparison object. It is not a second model that CKN must execute. Let $q: S \rightarrow \mathcal{X}$ identify the task state represented by the actual running state. The conditions in Section 5 connect this state to both the actual transition and the actual emitted observation.

4.2. Authority and displacement

The protected authority is the initialized selection of the declared task operation. Operation identity is extensional and domain-relative: it concerns the specified task transition and output law, not the identity of internal machinery. Substituting an observationally equivalent implementation is not displacement. A task trace departing from the reference trace on the same event history fails exact preservation. Such a departure can result from operator substitution, but need not do so; exact preservation is deliberately a stronger sufficient requirement.

Conversely, identical outputs on one history do not establish that two operations are identical. Input values can make different operations coincide. The local identities in Section 5 concern every event at every state in the declared geometric region, rather than an isolated output comparison.

An arbitrary reference may itself contain an input-triggered change of behavior. Preserving that reference reproduces that behavior. Therefore the theorem establishes resistance to substitution of the **specified operation**; it does not certify an undeclared instruction hierarchy inside that operation. This limitation cannot be repaired by attaching a privileged label to a reference model.

4.3. Non-degeneration and capability scope

Fix the decoding policy and the law of its exogenous randomness. For each prescribed finite admitted task-input history a , let $\mathcal{L}_p(a)$ be the law of the actual task-output trace under initialization p . A nontrivial instance must demonstrate both of the following on the identified task observation:

- For some fixed p , two equal-length admitted task-input histories a, a' satisfy $\mathcal{L}_p(a) \neq \mathcal{L}_p(a')$.
- For some $p \neq p'$ and one common admitted task-input history a , $\mathcal{L}_p(a) \neq \mathcal{L}_{p'}(a)$, as specified by the independently declared initialized task computations.

For deterministic systems these laws are point masses, so the conditions reduce to distinct output traces. Equal input-history lengths prevent differing observation horizons from supplying the distinction.

Variation caused only by decoder randomness does not demonstrate input responsiveness. These are comparisons under prescribed inputs, not conditional distributions selected from an adaptive run. The second condition does not require different transition formulas at a common state: an initialized task state may itself encode the selected operation.

Some admitted input differences must affect the claimed computation. Every submitted token need not survive admission, every admitted input need not change the response, and equivalent or irrelevant inputs may have the same effect. A system that ignores all admitted task input fails the first condition.

The observation must cover the computation for which capability retention is claimed. A retained input log, unused simulation, or auxiliary label does not establish preservation of an unobserved task. If the claimed distinction concerns only a header, the authority claim concerns that header. It cannot be extended to the accompanying content.

These distinctions exclude particular degeneracies; they do not define language competence. The stronger capability result is preservation of the actual reference task traces. For full-output claims, Y_Θ must include the full emitted output. A lossy observation supports only the property it observes. A baseline that is itself degenerate cannot establish a substantive capability claim merely by being preserved.

5. Geometric and operational conditions

5.1. Geometry with a specified interpretation

Choose a metric space $(\mathcal{Z}, \delta)$, a geometric observable $g: \mathcal{S} \rightarrow \mathcal{Z}$, an anchor $z_p \in \mathcal{Z}$, and a radius $a_p > 0$. Define

$$v_p(\Sigma) = \delta(g(\Sigma), z_p), \quad R_p = \{\Sigma : v_p(\Sigma) < a_p\}.$$

The scalar v_p measures geometric deviation, not task quality. The associated margin $a_p - v_p$ is derived from this specified geometry. It has a privilege interpretation only through the operational conditions below.

The map g need not be an orthogonal projection or a separately implemented component. It may overlap with, or equal, the task-state map q when their codomains permit. Changes in task state need not be small. Inputs may affect g directly and through recurrent task computation. No embedding-span exclusion is imposed.

5.2. Local hypotheses

For each declared p , fix constants $0 \leq \lambda_p < 1$ and $d_p \geq 0$. Require:

Initialization.

$$v_p(\Sigma_0) < a_p, \quad q(\Sigma_0) = x_0^p.$$

Geometric evolution. For every $\Sigma \in R_p$ and every $e \in \mathcal{E}$,

$$v_p(F_\Theta(\Sigma, e)) \leq \lambda_p v_p(\Sigma) + d_p.$$

This is a local-in-time condition on the full transition over a declared region. It does not assume the desired bound over actual histories. The residual d_p covers every effect permitted by the event, including decoder-dependent feedback. It is not inferred from raw token norms, and the inequality does not assume an additive decomposition of the model into privileged and input forces.

Operational connection. On the same region and for every event,

$$q(F_\Theta(\Sigma, e)) = B_p(q(\Sigma), e), \quad Y_\Theta(\Sigma, e) = H_p(q(\Sigma), e).$$

These identities connect the geometry to the actual task computation. They are obligations to establish for a candidate, not consequences of its weights being builder-controlled. A positive register unconnected to task execution cannot establish them for a task that is being overridden.

Strict geometric budget. Define

$$M_p = \max \left\{ v_p(\Sigma_0), \frac{d_p}{1 - \lambda_p} \right\} < a_p.$$

All objects and constants are fixed for the configuration and domain before applying the theorem. A different constant for each observed failure would be a different claim. No exact numerical value is needed for the general theorem; Section 7 supplies a numerical witness.

6. Persistence and preservation

Theorem 6.1. Geometric persistence

Under the initialization, geometric-evolution, and strict-budget conditions, every declared event history satisfies

$$v_p(\Sigma_t) \leq \lambda_p^t v_p(\Sigma_0) + d_p \sum_{j=0}^{t-1} \lambda_p^j \leq M_p < a_p.$$

The sum is zero at $t=0$. Thus the run remains in R_p at every finite step, for arbitrary history length.

Proof. The initialized state is in R_p . Assume the bound holds at step t . The state is then in the domain of the one-transition inequality, which gives the displayed bound at $t+1$. The geometric sum equals $(1 - \lambda_p^t)/(1 - \lambda_p)$, so the bound is a convex combination of the initial deviation and $d_p/(1 - \lambda_p)$, and is at most M_p . This closes the induction.

The inequality bounds deviation from a fixed privileged anchor; it does not assert pairwise contraction of states or a bound on differential gain from input. It does not contract every working-state trajectory, or collapse different privileged initializations together. In the disturbance-free case, deviation decays geometrically. Under continued disturbance, the bound gives a uniform envelope and an asymptotic upper bound of $d_p/(1 - \lambda_p)$; actual deviation can be smaller or zero.

Theorem 6.2. Preservation of the initialized task computation

Add the operational connection in Section 5.2. For the reference and actual system driven by the same event history,

$$q(\Sigma_t) = x_t, \quad y_{t+1} = y_{t+1}^p$$

at every finite step. Consequently all finite task-output traces are identical to those of the initialized reference operation.

Proof. Task states agree at initialization. Theorem 6.1 keeps the actual state in R_p . The two operational identities then give equality of the next task state and emitted task observation whenever the current task states agree. Induction proves the result.

This preserves whatever properties the declared reference task traces possess. It does not evaluate them. When the observation is the full emitted output, no quality threshold is needed to establish exact output preservation. If the reference performs distinct task computations, the protected system executes those same computations; it cannot replace them with a constant response and still satisfy the result.

6.3. What the hypotheses contribute

Geometric persistence alone would permit an irrelevant stable register. Task matching inside a region alone would permit inputs to drive the run out of that region. Theorem 6.2 combines the two. The operational identities are independent local task-fidelity hypotheses; geometric persistence keeps execution within their domain of validity but does not derive them from weight ownership, initialization, or distance alone. Section 7 establishes their simultaneous satisfiability, rather than leaving the conjunction as an uninstantiated premise.

These are standard induction and simulation arguments, not new general verification algorithms.

Establishing the local hypotheses may be difficult. No claim is made that their verification is always easier than trajectory analysis. Their role is to state independently checkable mathematical obligations for explicit systems and to derive the history-level result from them.

The operational identities imply that states in R_p with equal q values have equal next task states and outputs under the same event. This does not place g outside the task state, require nontrivial fibers of q , or make geometry observationally irrelevant. With $q = g$ equal to the identity, the equal- q statement simply compares a state with itself. Proposition 7.4 gives a nonconstant shared-state instance. A product decomposition or a separate authority register is not a general hypothesis of the theorem.

Geometry that factors through task state. Suppose $g = h \circ q$ on S , for a declared map $h : \mathcal{X} \rightarrow \mathcal{Z}$. Under the operational identities, geometric evolution is equivalent to

$$\delta(h(B_p(x, e)), z_p) \leq \lambda_p \delta(h(x), z_p) + d_p, \quad x \in q(R_p), e \in \mathcal{E}.$$

Indeed, substituting $g = h \circ q$ and $q(F_\Theta(\Sigma, e)) = B_p(q(\Sigma), e)$ gives the two sides directly, and every $x \in q(R_p)$ has a representing state in R_p . Once the factorization and represented domain are fixed, the inequality is a check on the reference. Establishing the factorization, represented domain, and operational connection remains part of the candidate-specification relation; the other hypotheses of Section 5 still apply.

Theorem 6.1 then bounds $\delta(h(x_t), z_p)$ by M_p . This confines the selected task observable. It does not by itself establish a loss of task capability or a cost relative to a baseline. A general geometric observable need not factor through q , and no decomposition into independent task and residual geometric constraints is inferred.

At equality $M_p = a_p$, the stated strict certificate does not apply. Failure of this sufficient inequality is not itself an override proof. A run may remain inside a region while approaching its boundary. Finite-horizon or probabilistic relaxations require separately stated claims.

7. Joint non-vacuity

Proposition 7.1. Abstract preservation for arbitrary task transducers

Let $p \in \{-1, +1\}$. Take any pair of total reference task transducers (B_p, H_p) over a common task state and event space, and a function $\psi(x, e)$ with $|\psi(x, e)| \leq 1$. Define a single fixed abstract transition system with state (r, x) , initialized at (p, x_0^p) .

Let $\sigma(r) = +1$ for $r \geq 0$ and $\sigma(r) = -1$ for $r < 0$. One transition is defined by

$$r' = \frac{1}{2}r + \frac{1}{2}\sigma(r) + \frac{1}{4}\psi(x, e), \quad k = \sigma(r'),$$

$$x' = B_k(x, e), \quad y' = H_k(x, e).$$

Then the actual task trace equals that of (B_p, H_p) for every declared event history. No condition of contraction, bounded dimension, or limited task complexity is imposed on the reference task dynamics.

Proof. Use $g(r, x) = r$, $z_p = p$, $a_p = 1$, and $v_p(r, x) = |r - p|$. In R_p , $\sigma(r) = p$, so

$$|r' - p| \leq \frac{1}{2}|r - p| + \frac{1}{4} < \frac{3}{4} < 1.$$

Hence $\sigma(r') = p$ and the actual task transition and output are precisely B_p, H_p . The constants are $\lambda_p = 1/2$, $d_p = 1/4$, initial deviation zero, and $M_p = 1/2 < 1$. Theorems 6.1 and 6.2 apply. In particular,

$$|r_t - p| \leq \frac{1}{2}(1 - 2^{-t}) < \frac{1}{2}.$$

Inputs and task state can affect the authority coordinate through ψ ; the authority coordinate selects the actual task transition. Thus the proof does not rely on an absent path of influence. The numerical instance below uses nonzero input coupling explicitly. Privileged configuration is supplied only at initialization, not injected anew at each step.

This is an existence proof in a permissive class of abstract transition maps. The state tuple and piecewise function are mathematical devices for demonstrating consistency. They do not prescribe a neural architecture, a training procedure, or a method of constructing weights. In particular, exact branch selection in this abstract map has not been established for a language model.

The arbitrary-transducer statement matters: the preservation result is not restricted to the capability of the scalar illustration below. It can inherit any specified reference computation. It also inherits any behavior already contained in that reference; the authority limitation in Section 4.2 continues to apply.

7.2. A task with no auxiliary authority label

Take $e = u \in [-1, 1]$, $\psi(x, u) = u$, and the reference task operation

$$B_p(x, u) = pu, \quad H_p(x, u) = pu, \quad x_0^p = 0.$$

The actual emitted task value is $y' = \sigma(r')u$. Changing p changes the numerical operation applied to the same input; changing u changes the actual task output. The protected variable is not an unrelated label, and the task computation is not an unused simulation. The input-to-authority coupling is nonzero: at fixed r, x , its coefficient is $1/4$.

Both public root values $+1$ and -1 are permitted subsequent inputs. Arbitrarily many copies are admitted. Proposition 7.1 gives $y_{t+1} = pu_t$ for every history, including histories chosen to displace the root.

This numerical operation is not a demonstration of language competence. It is an explicit witness that the mathematical requirements admit simultaneous task variation, privileged control of that same task, and resistance to accumulated nonprivileged perturbation. Capability inheritance for more general references follows from Proposition 7.1, conditionally on those declared references.

7.3. Strict improvement against a declared comparator

Consider an unprotected comparator with the same initialized state and task readout, but authority dynamics

$$r_{t+1}^0 = r_t^0 + \frac{1}{4}u_t, \quad y_{t+1}^0 = \sigma(r_{t+1}^0)u_t.$$

Starting with $p = +1$, supply $u_t = -1$ repeatedly. For the comparator, $r_t^0 = 1 - t/4$. For the protected system,

$$r_t = \frac{1}{2} + 2^{-(t+1)}.$$

After five inputs, the comparator has $r_5^0 = -1/4$ and emits $y_5^0 = +1$. The protected system has $r_5 = 33/64$ and emits $y_5 = -1$, preserving the initialized operation on that same input history.

The fixed- p reference establishes task preservation. The unrestored comparator establishes the resistance improvement. These are different comparisons. We can also give an explicit capability comparison against that same unrestored system, without weakening the attack domain.

Fix the comparison workload to all finite input histories whose every prefix satisfies

$$\left| \sum_{j=0}^{t-1} u_j \right| \leq 2.$$

On this workload, $r_t^0 = p + \frac{1}{4} \sum_{j < t} u_j$ remains within $1/2$ of p . Thus both systems emit precisely pu_t at every step. The workload is specified by an input condition, not selected from the protected system's results. It includes arbitrary first inputs in $[-1, 1]$ and arbitrarily long varying sequences. The repeated opposing input remains in the full resistance domain $[-1, 1]^*$ and produces the comparator failure above.

This is a single abstract comparator with exact preservation on a declared nontrivial workload and strict resistance improvement on an admitted attack. The protected system additionally preserves the reference task on the entire input domain. The unrestored integrator supplies a concrete existence comparison, not a representative language-model baseline. It is not evidence that CKN improves an arbitrary existing model, or that preserving an arbitrary reference removes vulnerabilities already inside it.

Proposition 7.4. Shared geometry and task state

The same-state case is also nontrivial. Set $\mathcal{S} = \mathcal{X} = \mathcal{Z} = \mathbb{R}$, use ordinary distance, and take $q = g$ equal to the identity. Let $p \in \{-1, +1\}$, $u \in [-1, 1]$, $r_0 = x_0^p = p$, $z_p = p$, and $a_p = 1$. Independently specify the reference task computations

$$B_p(x, u) = H_p(x, u) = \frac{1}{2}x + \frac{1}{2}p + \frac{1}{4}u.$$

Define one fixed actual system, with the same sign convention as Proposition 7.1:

$$F(r, u) = Y(r, u) = \frac{1}{2}r + \frac{1}{2}\sigma(r) + \frac{1}{4}u.$$

Here the emitted task observation is the updated state itself: $y_{t+1} = r_{t+1}$. There is no separate task coordinate or unobserved authority register.

Proof. On $R_p = \{r : |r - p| < 1\}$, $\sigma(r) = p$. Both operational identities therefore hold, and

$$|F(r, u) - p| \leq \frac{1}{2}|r - p| + \frac{1}{4}.$$

Thus $\lambda_p = 1/2$, $d_p = 1/4$, and $M_p = 1/2 < 1$. Theorems 6.1 and 6.2 apply. More explicitly, induction gives

$$r_t = p + \frac{1}{4} \sum_{k=0}^{t-1} 2^{-(t-1-k)} u_k, \quad |r_t - p| \leq \frac{1}{2}(1 - 2^{-t}).$$

At the first step the actual output is $p + u_0/4$, so changing admitted input or privileged initialization changes the actual task output. Within each region, the derivative of Y with respect to r is $1/2$ and with respect to u is $1/4$. Perturbations of the protected geometry therefore affect the emitted computation; their effects also persist into later steps through the displayed sum. The initialized regime remains preserved.

This example establishes a responsive computation whose protected geometry is its actual task state. It is a scalar shared-state instance. Proposition 7.1 separately permits arbitrary task transducers in a larger

abstract state space. These two witnesses do not jointly prove arbitrary task capability within a shared scalar geometry, or any language-model realization.

8. Capability tradeoffs without an output-quality oracle

The objective allows capability costs; it excludes collapse as a successful endpoint. Exact task preservation is a strong sufficient special case, not a claim that every useful realization must preserve every token of an existing model.

The mathematical witness has zero task-trace loss relative to its fixed- p reference while improving resistance over its unrestored comparator. Thus the abstract class contains systems that jointly preserve the declared task and improve resistance relative to the stated comparator. This establishes no unavoidable cost curve and no attainable tradeoff for learned models.

A practical comparison must name its baseline, task domain, and actual observations. The baseline cannot be reconstructed from the protected system after a failure. A domain of ordinary task comparisons must be fixed independently of where the protected system happens to perform well. Changed behavior on an operator-substitution attempt is the resistance gain, not automatically a capability loss.

A capability comparison may use a separate baseline and a numerical tolerance without weakening the exact authority contract. Appendix B gives the complete elementary propagation bound for that comparison. Its metric, domain, and readout assumptions must be declared. A small latent-state norm alone does not establish comparable linguistic coherence; a discontinuous decoder can turn a small internal change into a different emitted token. Relaxing exact matching to the protected operation itself would be a different authority claim.

The paper does not specify a universal acceptable score, an absolute notion of better output, or a training objective. It proves an exact preservation case and identifies what a broader tradeoff claim would need. A claim of useful CKN realization must establish resistance and retained capability in the same system; separate witnesses for each are insufficient.

9. Meaning, limitations, and prior art

The surviving mathematical claim is substantive and limited: local geometric stabilization plus a local connection to an independently specified task operation yields persistent authority over that operation and preservation of its task traces; the conjunction has explicit nontrivial instances.

The proof does not establish authority over arbitrary natural-language instructions. It does not extract a task specification, a meaningful geometric observable, or a disturbance bound from arbitrary weights. Parameter control identifies the builder's control surface, not a proof of a suitable neural realization. Nothing here requires a particular learning method.

The mathematical tools are established. Safety verification relates initial regions, excluded regions, and evolution conditions [1]. Contraction analysis supplies the familiar comparison between restoration and disturbance [2]. Simulation relations connect one-transition matching to execution-level observation preservation [3]. The short arguments are reproduced so their precise scope is inspectable. No novelty claim is made for those mathematical tools.

CKN's proposed contribution is their joint use to specify initialized task authority, nonzero permitted coupling, and preservation of the computation actually emitted. Whether this formalization captures useful forms of privileged model conditioning, and whether suitable weights can realize the intended resistance-capability tradeoff, remain open research questions.

10. Release-candidate scope

Version 1.4.2 is a release candidate for the stated sufficient theory. It derives a trajectory bound from individual-transition conditions, proves preservation of actual task traces, and establishes joint non-vacuity with an explicit attack comparison. The shared-state witness shows that the conditions do not require geometry to sit outside task computation. Admitted-input responsiveness, endogenous privileged-token stabilization, and the limits of the geometric interpretation are explicit in the formal scope.

The broad goal remains resistance to adversarial override with retained generative competence and an acknowledged capability tradeoff. The proved result concerns specified task-operation authority. General language-model realization and the bridge from ordinary privileged instructions to that formal authority have not been established.

References

- [1] Prajna, S., and Jadbabaie, A. (2004). *Safety Verification of Hybrid Systems Using Barrier Certificates*. HSCC 2004, LNCS 2993, 477-492. [Author manuscript](#).
- [2] Lohmiller, W., and Slotine, J.-J. E. (1998). *On Contraction Analysis for Non-linear Systems*. Automatica 34(6), 683-696. [Author manuscript](#).
- [3] Girard, A., and Pappas, G. J. (2011). *Approximate Bisimulation: A Bridge Between Computer Science and Control Theory*. European Journal of Control 17(5-6), 568-578. [Author manuscript](#).

Appendix A. Claim ledger

Claim	Status
Span exclusion alone prevents subsequent influence	Rejected; two-dimensional counterexample.
A positive scalar alone establishes privilege	Rejected; operational connection is required.
Local geometric hypotheses imply persistence	Proved in Theorem 6.1.
Persistence plus task matching preserves actual task traces	Proved in Theorem 6.2.
Both can hold with nonzero input coupling and nontrivial tasks	Proved in Proposition 7.1 and instantiated in Section 7.2.
Protected geometry can coincide with responsive task state	Proved in Proposition 7.4; scalar shared-state instance.
Resistance can improve without task collapse in the abstract class	Explicit comparator and task witness in Section 7.3.
Endogenous privileged tokens may sustain the initialized regime	Permitted within the complete running dynamics; no realization proved.
Input or context limits alone establish persistent stabilization	Not implied; the full transition inequality remains required.
All privileged natural-language instructions have this formal meaning	Not established.
Suitable neural weights exist or can be obtained	Not established.
Comparable language quality and a useful practical tradeoff are achieved	Not established.

The companion *CKN v1.4.2 - Mathematical Audit and Revision Disposition* records additional proof checks, adversarial witnesses, and revision history. It is supplementary; the definitions and proofs supporting this paper's claims are contained here.

Appendix B. Optional approximate capability comparison

This lemma supplies a separate comparison of actual observed output to a baseline. Retain the exact conditions establishing authority over the protected operation. Independently declare a capability baseline B^0, H^0 , which may differ from that operation, and a task representation $Q: S \rightarrow \mathcal{X}^0$. Let $(\mathcal{X}^0, \rho)$ and $(\mathcal{Y}, \Delta)$ be metric spaces. The baseline maps are defined on all of $\mathcal{X}^0 \times \mathcal{E}$, with B^0 taking values in $\mathcal{X}^0$. From a declared $x_0^0 \in \mathcal{X}^0$, under the same event history, write

$$x_{t+1}^0 = B^0(x_t^0, e_t), \quad y_{t+1}^0 = H^0(x_t^0, e_t).$$

Fix constants $\eta, \nu, L, K \geq 0$. Require the first and third inequalities below for every $\Sigma \in R_p$ and $e \in \mathcal{E}$, and the second and fourth for every $x, \tilde{x} \in \mathcal{X}^0$ and $e \in \mathcal{E}$:

$$\rho(Q(F_\Theta(\Sigma, e)), B^0(Q(\Sigma), e)) \leq \eta,$$

$$\rho(B^0(x, e), B^0(\tilde{x}, e)) \leq L\rho(x, \tilde{x}),$$

$$\Delta(Y_\Theta(\Sigma, e), H^0(Q(\Sigma), e)) \leq \nu,$$

$$\Delta(H^0(x, e), H^0(\tilde{x}, e)) \leq K\rho(x, \tilde{x}).$$

All constants are uniform over the stated domains. Since Q takes values in $\mathcal{X}^0$ and the total baseline transition preserves that space, the global baseline bounds cover both represented and baseline states at every finite step. Theorem 6.1 keeps the actual state in R_p , where the mismatch bounds apply. These are additional comparison hypotheses, not consequences of geometric persistence; no separate trajectory-domain assumption is required for this comparison.

Lemma B.1. Define $\varepsilon_t = \rho(Q(\Sigma_t), x_t^0)$. Then

$$\varepsilon_t \leq L^t \varepsilon_0 + \eta \sum_{j=0}^{t-1} L^j, \quad \Delta(y_{t+1}, y_{t+1}^0) \leq K\varepsilon_t + \nu.$$

Proof. Insert $B^0(Q(\Sigma_t), e_t)$ between the actual and baseline next task states. The triangle inequality and the first two hypotheses give $\varepsilon_{t+1} \leq \eta + L\varepsilon_t$. Induction gives the displayed geometric sum, with empty sum zero at $t=0$. Insert $H^0(Q(\Sigma_t), e_t)$ between the actual and baseline outputs and apply the last two hypotheses to obtain the output bound.

For $L < 1$ the task-state error has a uniform upper bound. For $L = 1$ the accumulated term is $t\eta$; for $L > 1$ it may grow geometrically. A finite-horizon estimate is not an indefinite preservation guarantee. The metric and readout conditions carry the comparison's meaning; they do not supply a universal language-quality score. If Δ is the discrete output metric, a strict bound below 1 entails exact observation equality, while larger bounds can be uninformative. A discontinuous decoder may invalidate a proposed Lipschitz readout condition.

The protected operation's exact authority contract remains unchanged. Replacing its own operational identities by approximate ones would require a separately stated authority theorem.